

Effects of Fitness Advertising on Weight and Body Shape Dissatisfaction, Social Physique Anxiety, and Exercise Motives in a Sample of Healthy-Weight Females

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The effects of exposure to fitness advertising on multiple dimensions of female body image were explored. Healthy weight females ($N = 185$) were randomly assigned to a model-focused, product-focused, or control group and completed body image inventories during a pre-test and following exposure to fitness advertising 1 week later. There were no significant main effects for group or time on any body image measures. A group by time interaction was observed for affective body image, $F(1, 179) = 45.52, p < .001, \eta^2 = .32$. Females exposed to model-focused fitness advertising reported higher social physique anxiety compared with females in the other groups. These findings suggest that multidimensional body image perspectives are informative, especially when examining the effects of media advertising.

Body image is a multidimensional construct consisting of body-related cognitions, affect, and behavior (Bane & McAuley, 1998; Cash & Pruzinsky, 2002). In this regard, body image refers to how individuals think (a cognitive dimension), feel (an affective dimension), and behave (an investment or behavioral dimension) with respect to their physical attributes of weight, body shape and size, and appearance (Cash, 1994a; Cash & Pruzinsky). Cognitive body image reflects an evaluative component and refers to levels of satisfaction with one's physical self (Cash; Cash & Pruzinsky, 1990; Muth & Cash, 1997). Cognitively based evaluations, which are commonly reported as discrepancies between one's current physical state and an ideal or desired state, tend to be derived from internalizing ideals that are often represented in the media (Cash & Pruzinsky, 1990, 2002). The affective body image domain refers to unique emotional experiences associated with cognitive evaluations and internalized standards in specific contexts (Cash, 1994b; Cash & Pruzinsky, 2002). Body-related affect usually

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involves measures of specific emotions such as anxiety or shame. Finally, the behavioral domain of body image is primarily focused on actions that result from or in reaction to appearance and body perceptions (Bane & McAuley, 1998). According to Cash (1994a), behavioral body image refers to an investment to manage and enhance one's physical characteristics.

Women are more likely to report body image concerns in all domains compared with men (Cash & Pruzinsky, 2002; Garner, 1997; Hausenblas, Janelle, Gardner, & Hagan, 2002; Martin, Rejeski, Leary, McAuley, & Bane, 1997). According to social-cultural perspectives, western societal standards emphasize thinness and body shapes that may be genetically impossible for many women to achieve through healthy means (Thompson, Heinberg, Altabe, & Tantleff-dunn, 1999). In spite of the impossible standard, the ideal figure is accepted and internalized by many women (Tiggemann & McGill, 2004). This internalization represents a cognitive evaluation, elicits affect, and supports investment into the management of appearance and weight. It is therefore important to understand the impetus underlying the development and subsequent internalization of female body image ideals.

Fashion media are one of the strongest transmitters of unrealistic body shape and appearance ideals, and have been consistently linked to widespread body image and weight concerns among women in western society (Groesz, Levine, & Murnen, 2002; Posovac, Posovac, & Posovac, 1998; Thompson et al., 1999; Tiggemann & McGill, 2004). Limited research has examined other domains, such as exercise and fitness. While fashion advertising focuses on tall, thin and tubular attractive female body shapes (Groesz et al.), exercise and fitness advertising likely contributes an additional body ideal in highlighting muscle tone. Limited evidence suggests muscle tone is desired by women (Garner, 1997). Thus the characteristics of increased muscularity and tone, as portrayed in fitness media, could influence females' cognitive, affective, and behavioral body images.

Several researchers examining the effects of advertising on body image have used Festinger's (1954) social comparison theory (Cattarin, Thompson, Thomas, & Williams, 2000; Heinberg & Thompson, 1995; Posovac et al., 1998; Tiggemann & McGill, 2004). Based on the theory, individuals have inherent tendencies to evaluate themselves through comparisons with others. When women compare themselves with the physically attractive thin (ideal) figures portrayed in the media, it is often an upwards comparison in which they lack the attributes portrayed and this fosters negative mood and body image (Cattarin et al.; Festinger; Gulas & McKeage, 2000; Tiggemann & McGill). What is unclear is the extent to which the effects of media images differentially yet simultaneously influence the multiple (i.e., cognitive, affective, and behavioral) dimensions of body image, and if different advertising media (i.e., fitness) impact these multidimensional aspects of body image. It is important to explore the possible effects of exposure to fitness advertising given the increase in this

type of advertising (Yoon & Choi, 2005). This ever-increasing prevalence of advertising depicting fit and toned female bodies may create a body image context that mirrors the fashion industry, whereby exposure to models in advertising leads to more negative body image compared with exposure to nonmodel-based advertisements. This proposition requires further investigation. There has been limited exploration of the impact of sports media on women's body image characteristics. Specifically, sport magazine reading predicted greater body satisfaction among older adolescents (Harrison & Fredrickson, 2003), and sports television was negatively correlated with adolescent female's body dissatisfaction (Tiggemann & Pickering, 1996). Furthermore, Harrison (2000) reported that regular exposure to sports print media was directly linked to negative body image. Combined, these studies highlight important associations between sports media and body image, and the inconsistencies in the direction of effects needs to be better understood using alternate modes of media and distinct samples.

The general purpose of this study was to determine whether exposure to fitness-related magazine advertisements contributed to body image disturbances among healthy-weight females. More specifically, differences between exposure to model-focused fitness advertising and product-focused fitness advertising were examined and compared with a no-media exposure control group. To address the purpose of this study, we assessed cognitive body image using a Figure Rating Scale (FRS; Stunkard, Sorenson, & Schulsinger, 1983). This measure was developed to target individuals' evaluations of their current and ideal body shapes and is often used as an indicator of dissatisfaction (Bane & McAuley, 1998; Banfield & McCabe, 2002). In line with the definition of cognitive body image, scores on the FRS reflect self-perceived discrepancies in body shapes. Affective body image was assessed as social physique anxiety (Hart, Leary, & Rejeski, 1989), which is the body-related concerns associated with perceptions that others are negatively evaluating one's figure. Social physique anxiety (SPA) is a commonly reported construct used to assess affective antecedents and outcomes in physical activity and health research (Kowalski, Mack, Crocker, Niefer, & Fleming, 2006; Martin Ginis & Leary, 2004; Sabiston, Crocker, & Munroe-Chandler, 2005). While most of the literature examines trait SPA, there is growing research identifying contextual features (Carron & Prapavessis, 1997; Kowalski et al. 2006; Sabiston, Sedgwick, Crocker, Kowalski, & Mack, 2007). It is important to continue to examine contextual features of SPA, such as possible fluctuations as a result of internalizing fitness media. Finally, behavioral body image was assessed predominantly as the weight and appearance reasons that women identify for exercising. According to researchers (Kowalski et al.; Sabiston et al.), exercise is a common strategy that women use to manage and improve their appearance and weight. The Reasons for Exercise Inventory (Silberstein, Striegel-Moore, Timko, & Rodin, 1988) was

used to capture these reasons, as well as fitness/health, stress relief, and socializing exercise motives. Since actual exercise behavior following the media exposure was not measureable within the context of this study, the authors felt that motives for exercise (which are indicators of behavioral investment) could be manipulated. It was hypothesized that females exposed to model-focused fitness advertising would report more detrimental changes in cognitive, affective, and the weight and appearance behavioral body image domains compared with females exposed to product-focused fitness advertising or a control group. Product-focused advertising was not expected to influence body image scores.

Method

Participants

Female undergraduate students from a Southern Ontario university were asked to participate in this study by completing a self-report survey twice during class. Two hundred and eleven students completed the pretest, and, 87.7% also completed the posttest and met the eligibility criteria. Given the known heightened body image scores for underweight and overweight women (Crawford & Eklund, 1994; Fitzgibbon, Blackman, & Avellone, 2000; O'Dea & Caputi, 2001; Pesa, Syre, & Jones, 2000), the current study focused on healthy-weight women with body mass index (BMI) values of 18.5–25 kg/m² (Ogden et al., 2006). This final sample ($N = 185$) reported a mean age of 20.12 ($SD = 1.70$) years, mean BMI of 21.36 ($SD = 2.12$) kg/m², and the majority (86.4%) reported exercising at least once per week.

Measures and Materials

A number of scientifically supported instruments were used to assess the multidimensional nature of body image. Also, BMI was calculated from self-reported height and weight for each participant by dividing their weight (kg) by their height (m) squared.

Cognitive body image. Participants were presented with the Figure Rating Scale (Stunkard et al., 1983). The Figure Rating Scale measures satisfaction with body shape using a 9-point continuum of female body figure diagrams ranging from extremely thin to extremely overweight. Participants were asked to indicate the body figure that approximates their current weight and the figure they considered ideal. Consistent with previous research and propositions (Bane & McAuley, 1998; Banfield & McCabe, 2002; Cash & Pruzinsky, 2002), the absolute difference between the current and ideal ratings was used as a measure of cognitive body image.

Affective body image. The Social Physique Anxiety Scale (Hart et al., 1989; Martin et al., 1997) is a 9-item self-report inventory that measures social physique anxiety as an affective dimension of body image (Bane & McAuley, 1998). Participants were asked to indicate the degree to which statements (e.g., *It would make me uncomfortable to know others were evaluating my figure*) are characteristic of them on a 5-point Likert type scale, ranging from (1) *not at all* to (5) *extremely*. Mean scores were calculated for this scale, with higher scores indicating greater body-related anxiety. Given emerging research examining the state-like attributes of social physique anxiety (Kowalski et al., 2006; Sabiston et al., 2007; Van Raalte, Cunningham, Cornelius, & Brewer, 2004), we used this scale as an indicator of state affective body image. The scale has adequate ($\alpha > .85$) psychometric properties with females (Crawford & Eklund, 1994; Sabiston et al., 2005).

Behavioral body image. The Reasons for Exercise Inventory (Silberstein et al., 1988) is composed of 24 items that assess the importance of exercise motives that are rated from (1) *not at all important* to (7) *extremely important*. Research evidence (Cash, Novy, & Grant, 1994; Sabiston et al., 2005) supports a four-factor scale structure, including exercise motives of weight and appearance (e.g., *to improve my overall body shape*), fitness and health (e.g., *to improve my endurance and stamina*), stress and mood (e.g., *to improve my mood*), and socializing (e.g., *to meet new people*). Previous research has suggested that the weight and appearance subscale is highly associated to other measures of body image (Sabiston et al.). The reasons for exercising were assessed as an indicator of behavioral management strategies that women use in response to their body images.

Advertising manipulation. For the experimental manipulation, all athletic/fitness footwear advertisements were selected from a 1-year period of common female fitness magazines (*Shape*, *Fitness*, and *Self*). The focus on athletic footwear was simply to maintain consistency in the advertisements. To obtain a final sample of fitness advertisements, the first author and 11 graduate students coded all advertisements for model and product focus. Specifically, the coders were asked to confidentially rate the advertisements based on the degree to which each projected a model or a product (athletic shoes) by answering three questions: (a) *To what degree is this advertisement focused on a human model* (1 = *very model-oriented* to 5 = *not at all model-oriented*); (b) *To what degree is this advertisement focused on athletic shoe(s)* (1 = *not at all athletic shoe-oriented* to 5 = *very athletic shoe-oriented*); and (c) *Rate this advertisement on the following scale*: (1) *very model focused* to (5) *very athletic shoe focused*. The 15 highest scores (product-focused) and lowest scores (model-focused) were selected and scanned into two Microsoft Office PowerPoint presentations. The final product-focused advertisements depicted only athletic shoes and no models. The separate presentations revealed a new advertisement every 15 seconds until the 15 advertisements had

been viewed. This timeframe of exposure was consistent with previous research (Heinberg & Thompson, 1995; Monro & Huon, 2005).

Procedures

Following university ethics board approval, introductions to the study protocol were made to female students from a variety of undergraduate courses. Information and consent forms were distributed to interested participants, who were informed that the study would involve the completion of two questionnaires (approximately one week apart) during class time. It was emphasized that participation in the study was voluntary and would not influence course grades. Following consent procedures, the pretest data were collected by having participants complete the body image questionnaires.

Approximately 1 week following the pretest, female students from the same undergraduate classes were asked to take part in the posttest. Participants within each targeted class were randomly assigned to be exposed to model-focused fitness advertisements, to product-focused fitness advertisements, or to no advertisements and were instructed to move to separate classrooms that were set up to present the manipulations. Participants in the advertising groups were exposed to the advertisements followed by the completion of the same instruments used during the pretest. Participants in the control group were asked to complete the questionnaire.

Data analysis. The data were screened for psychometric properties and tests of assumptions (Tabachnick & Fidell, 2001). Pearson correlations were examined for relationships among the body image measures. Preliminary analyses of variance (ANOVA) were conducted to examine group differences on the pretest scores. To test the hypothesis of within- and between-group differences on body image following the manipulation, three (group) by two (time) repeated measures MANOVAs with follow-up ANOVAs were used. In cases where significant interactions (i.e., manipulation effects) occurred, simple main effects were calculated to examine the significance of group changes.

Results

Descriptive Statistics

Means and standard deviations on the demographic variables and pretest body image scales are presented in Table 1 separately for the control ($n = 57$), model-focused ($n = 67$) and product-focused ($n = 61$) groups. Based on these descriptives and other measures of central tendency, the sample reported low levels of cognitive body image (median absolute values for the absolute body

Table 1

Means and Standard Deviations for Pre- and Posttest Descriptive Variables for the Total Sample, and Model- and Product-Focused Groups

	Scale range Score range	Control Means (SD)	Model focused Means (SD)	Product focused Means (SD)
Age (yrs)	— (18–27)	20.35 (1.69)	19.98 (1.39)	19.98 (1.57)
BMI (kg/m ²)	— (18.53–24.96)	21.03 (2.38)	21.35 (1.39)	22.16 (1.90) ^a
Body shape discrepancy	0–7			
Pretest	(0–3)	.72 (.68)	.75 (.60)	.90 (.77)
Posttest	(0–3)	.74 (.67)	.75 (.60)	.80 (.68)
Social physique anxiety	1–5			
Pretest	(1.44–5.0)	2.94 (.61)	2.95 (.80)	3.06 (.75)
Posttest	(1.36–5.0)	2.94 (.59)	3.28 (.80)	3.02 (.76)
REI—weight/appearance	1–7			
Pretest	(1.0–7.0)	4.39 (1.16)	4.48 (1.27)	4.77 (1.50)
Posttest	(1.0–7.0)	4.38 (1.16)	4.49 (1.34)	4.80 (1.49)
REI—fitness/health	1–7			
Pretest	(2.0–7.0)	5.04 (1.18)	5.05 (1.19)	5.07 (1.14)
Posttest	(2.0–7.0)	5.01 (1.17)	5.11 (1.16)	5.24 (1.16)
REI—stress/mood	1–7			
Pretest	(1.0–7.0)	4.73 (1.13)	4.89 (1.27)	4.70 (1.32)
Posttest	(1.3–7.0)	4.79 (1.17)	5.02 (1.22)	4.67 (1.25)
REI—social	1–7			
Pretest	(1.0–7.0)	3.43 (1.75)	3.48 (1.73)	3.56 (1.61)
Posttest	(1.0–7.0)	3.27 (1.59)	3.54 (1.77)	3.60 (1.59)

BMI = body mass index; REI = reasons for exercise inventory.

^aBMI significantly higher for women in the product-only group compared with women in the control and model-only group. There were no other significant differences among pretest scores between the groups.

Table 2

Correlations among Cognitive, Affective, and Behavioral Body Image Pretest Measures

	1.	2.	3.	4.	5.	6.
Body shape discrepancy	—					
Social physique anxiety	.45**	.89				
REI-weight/appearance	.43**	.48**	.87			
REI-fitness/health	-.18*	-.02	.08	.88		
REI-stress/mood	-.09	-.07	-.01	.55**	.76	
REI-social	-.01	-.04	.01	.32**	.37**	.87

Note. Cronbach's alpha coefficients are presented on the diagonal.

REI = reasons for exercise inventory.

* $p < .05$. ** $p < .001$.

shape discrepancy was 1.0 across all groups). Levels of social physique anxiety were low to moderate (median scores ranged from 2.8 to 3.0 across the groups). Women reported highest scores on the fitness and health behavioral body image subscale (median values of 5.0–5.4 across the groups), followed by the weight and appearance motives (median values of 4.3–5.0 across the groups). Preliminary ANOVA revealed no significant group differences on pretest measures of age, or the cognitive, affective, and behavioral body image measures (see Table 1). Females in the product-focused group reported significantly higher BMI, $F(2, 179) = 4.71$, $p = .01$, compared with the control and model-focused groups. BMI was treated as a covariate in further analyses. Pearson correlations among the pretest body image measures revealed significant moderate inter-correlations (see Table 2).

Main Analyses

Posttest scores are reported in Table 1. To explore the hypothesis of group differences following exposure to fitness advertising, a Repeated Measures ANCOVA was conducted controlling for baseline BMI. There were no significant main effects for time, $F(1, 181) = .81$, $p = .56$, $\eta^2 = .03$ or group, $F(2, 179) = 1.08$, $p > .05$, $\eta^2 = .07$. The group by time interaction was significant, $F(2, 179) = 6.12$, $p < .001$, $\eta^2 = .17$. Univariate analyses revealed that the group by time interaction was significant for social physique anxiety, $F(2, 182) = 37.77$, $\eta^2 = .20$. Based on tests of simple main effects, the model-focused group increased significantly in affective body image, $F(1, 66) = 92.13$, $\eta^2 = .58$, while the product-focused, $F(1,$

60) = .54, $p > .05$, $\eta^2 = .01$, and control, $F(1, 56) = .03$, $p > .05$, $\eta^2 = .001$, groups did not change. There were no other significant group by time interactions, suggesting no differences on cognitive or behavioral body image for individuals in the model-focused or product-focused advertising groups or in the control group.

Discussion

The findings from this study suggest that fitness advertising provides unique influences on the various modalities of body image. Exploring the cognitive, affective, and behavioral modalities of body image among a sample of healthy weight females helps advance body image theory and highlights the importance of identifying the multidimensional components. Based on this study, it may be the affective body image domains that are primarily influenced by exposure to fitness advertising. The consideration of fitness advertising as a novel area of body image influence, expanding on the research with thin fashion model media, is timely given that sport and fitness models are increasingly used in advertising (Yoon & Choi, 2005). Furthermore, there is increasing interest in the broader domain of sports media (Harrison, 2000; Harrison & Fredrickson, 2003; Tiggemann & Pickering, 1996), for which this study lends some support.

The primary hypothesis stating that model-focused fitness advertising would increase reports of negative body image compared with product-focused fitness advertising was supported. There was a significant increase in body-related anxiety evident only following exposure to fitness model advertising images. Within the fashion industry and other forms of media, it has been established that women exposed to figures highlighting the body are more likely to report decreased body satisfaction (Groesz et al., 2002; Hausenblas & Fallon, 2002), increased general negative affect (Pinhas, Toner, Ali, Garfinkel, & Stuckless, 1999; Tiggemann & McGill, 2004), and anxiety specifically related to the body (Monro & Huon, 2005). The current findings suggest that model-focused fitness advertising has similar effects on these aspects of body image, where social physique anxiety (SPA) showed heightened levels following the advertising exposure. These findings lend support to the growing body of research indicating SPA, which has been predominantly explored as a trait, may be influenced by both dispositional and situational factors (Martin Ginis & Leary, 2004). For example, there is limited evidence that individuals report fluctuating body-related anxiety intensities that are influenced by the social context (Sabiston et al., 2007; Van Raalte et al., 2004) and by exposure to health-based (Berry & Howe, 2004) and model-focused (Monro & Huon) advertising. Framed within a larger body image context, SPA has demonstrated attributes supporting both trait and state foundations (Carron & Prapavessis, 1997; Kowalski et al., 2006; Martin Ginis & Leary; Sabiston et al., 2007; Van Raalte et al., 2004). The findings that SPA scores were stable across time in the control group, and changed

significantly for the model-focused group, may highlight the trait and state characteristics of SPA.

Self-objectification theory (Fredrickson & Roberts, 1997) may provide one way of explaining the findings linking exposure to model-focused media and affective body image. Specifically, the theory suggests that the constant focus on women's bodies and objectification of them leads women to monitor their outward appearance and body shape, to be concerned with others' perceptions of their appearance, and to be more preoccupied with what the body looks like rather than what it can do (Fredrickson & Roberts; Greenleaf, 2005). It is possible that exposure to these muscular and toned female figures primed self-objectification, and, according to the theory, led to heightened body shame and anxiety (which was assessed as social physique anxiety) compared with exposure to product-related advertising. Similar results were noted in a recent study with fashion models (Harper & Tiggemann, 2008). Implications of these studies suggest that self-objectification may be elicited in women who are exposed to media images, and heightened affective states are likely outcomes.

There were no significant group differences on cognitive and behavioral body image dimensions, yet these results may not be entirely unfounded. There is some research to suggest that media models representing an extreme body standard do not impact body image in the same way as those representing a more realistic female standard (Smeeters & Mandel, 2005). Thus, the toned and muscular models represented in fitness magazines may be seen as extreme and have little impact on these dimensions of body image. According to the premise of social comparison theory (Festinger, 1954), individuals are less likely to use comparative processes when they perceive the available comparative group or individual as divergent from themselves. If no comparison exists, there is no impact on cognition or behavior. Thus, a context in which fitness models are perceived to be extreme (different from the females in the sample) would not satisfy the condition of social comparison, and there would be no anticipated cognitive or behavioral outcomes. Based on these observations, future research should focus on understanding the extent to which fitness models are perceived to be attainable and realistic.

Another reason that might explain the nonsignificant effects of fitness media on cognitive and behavioral body image dimensions may be rooted in self-objectification theory (Fredrickson & Roberts, 1997). In contrast to fashion media, Harrison and Fredrickson (2003) have suggested that sports media may be focused on what the body can do rather than how it appears. This stipulation implies that there is a difference in the meanings of physiques presented in sports media compared with other thin-ideal media such as fashion advertising, and the meaningful differences may protect against self-objectification. While these differences in meanings have not been adequately studied, it is possible that females in the current study perceived the fitness media to present the

body as an instrument that is achieving rather than the body according to externally perceivable traits such as appearance. Previous research focused primarily on lean versus nonlean sports media has reported a positive association between sports media and body satisfaction (Harrison & Fredrickson). Body satisfaction was conceptualized to include constructs similar to cognitive and behavioral body image as measured in the current study. Thus, it could be that the meaning of the body in fitness media is unique and future research should explore this possibility, where qualitative methods would be useful to better understand experiences and meaning associated with fitness media.

Finally, it is possible that change in cognitive and behavioral body image was not observed due to the assessment of these dimensions. For example, the Figure Rating Scale (Stunkard et al., 1983) includes only nine body silhouettes on which individuals can indicate their current and ideal shapes. With limited distinctions among the shapes, it is not likely that women would significantly change their ideal figure rating. Also, these figures depict adiposity rather than muscle mass, and in this type of study changes in muscle tone would likely be informative. Thus, further measures associated with multiple aspects of body shape need to be explored (Furnham, Titman, & Sleeman, 1994; Thompson et al., 1999).

While not a main focus in the current study, it is worthwhile noting that the cognitive, affective, and behavioral (weight and appearance exercise motives) body image measures demonstrated positive moderate correlations ($r = .43-.48$) that fit within a multidimensional framework of body image (Bane & McAuley, 1998; Cash, 1994a). Cognitive body image was negatively related to fitness and health reasons for exercise, suggesting that individuals who report higher body shape dissatisfaction are less likely to exercise for fitness and health reasons and more likely to exercise as a way of managing or altering weight and appearance. These inverse relationships may infer the existence of a continuum of behavioral body image indicators, whereby weight reasons are at one extreme, and fitness and health motives are at the other end. This would be in line with the organismic integration theory (OIT: Deci & Ryan, 1985), whereby individuals engage in behavior for reasons that range from extrinsically endorsed (i.e., weight and appearance) to internalized and self-determined (i.e., fitness, health, enjoyment). Based on research with OIT (Deci & Ryan) principles and body image definitions (Cash, 1994b), women who internalize ideal physiques portrayed in media such as fitness magazines may experience heightened negative cognition and affect, and may feel they need to exercise for weight and appearance reasons rather than for fitness, health, stress and mood, or social reasons (which were all moderately positively correlated in the current study). In line with the current study findings, Frederick and Morrison (1996) found that university-aged participants with higher levels of SPA were more extrinsically motivated to exercise (e.g., weight

concerns), and also reported more frequent exercise behavior compared with participants with lower levels of SPA. Future research needs to focus on the actual behavioral outcomes of body image within a multidimensional framework.

With the advancement of studies investigating different methods of body image assessment, unique forms of media (e.g., health, fitness and sport, fashion), and controlling for distinct elements within the media (e.g., model and product focused), we will be able to better understand the factors that influence the multiple dimensions of body image. This study attempts to advance the understanding of media effects on body image by limiting the advertisements to athletic footwear, examining the effects of both product and model-related fitness advertisement elements, and conceptualizing body image as a multidimensional construct. In response to the call for more attention on this issue (Groesz et al., 2002), we have provided a well-controlled study to explore the impact of a unique form of media on body image. We were also able to identify a healthy-weight group of females that appear to be vulnerable to the effects of fitness advertising on their body images. However, it is important to note that this group of healthy-weight women was classified based on BMI scores only, which do not account for adiposity levels. As such, BMI was used as a proxy for health in the current study. There are additional study limitations that require attention. First, a larger sample size would enable further discrimination among women with preexisting body image issues within each experimental group. Second, the participants were recruited from classes, and there was no attempt to control for the possible clustering of responses. Third, we did not measure changes in actual behavior, such as exercise. It is possible that exposure to model- or product-focused advertising could lead to changes in exercise behavior intensity or duration that were not captured in this study. Finally, some of the measures used in the current study may preclude an assessment of change.

In spite of the limitations, the current study offers theoretical and practical implications for conceptualizing body image as a multidimensional construct and for understanding unique sociocultural determinants of the body image dimensions. Fitness advertising appears to influence body image affect in a comparable way to the effects of fashion advertising. Additionally, exposure to model-focused fitness media has differential effects on the various dimensions of body image. While fitness images depicting models were influential sources of more negative body image affect, these images did not appear to impact all healthy-weight females equally. Our findings suggest there is a range of susceptibility to body image change even within a sample of healthy-weight females. These differences in susceptibility, coupled with the changing landscape in model-focused advertising depicting greater muscularity, may lead to necessary changes in the understanding and managing of body image.

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